Laboratory 1: Quantum Random Number Generator

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ABSTRACT In this report we analyzed experimental data acquired during a laboratory session on Quantum Random Number Generators (QRNGs), focusing on the comparison between trusted and Source-Device-Independent (SDI) protocols. The experimental setup is based on a heralded single-photon source combined with polarization measurements, allowing the generation and characterization of different quantum states. Measurements are performed along Alice's side (measuring channel), while coincidence detections with Bob (the sender) are used to reduce background noise and validate single-photon events (verification channel). Time-tagged data detection are filtered by selecting an appropriate coincidence window over the temporal correlation histograms. Using the measured statistics, we estimated the min-entropy achievable under different randomness generation protocols: **(a fully trusted protocol, SDI protocols based on mutually unbiased bases, full state tomography, SDI protocol employing a square POVM)** We assessed the security of the generated randomness by applying a strong seeded randomness extractor based on a universal-2 family of hash functions. Using the Leftover Hashing Lemma, we study how the security parameter and the final length depends on the experimentally estimated min-entropy and on the extractor block size. Finally, we used a concrete implementation of a universal-2 family of hash functions to extract randomness from the collected raw data.

I. EXPERIMENTAL SETUP

A. GENERAL EXPERIMENTAL SETUP

All experimental runs share the same underlying optical and detection architecture; therefore, the general setup is described once in this section.

Photon pairs are generated through spontaneous parametric down-conversion (SPDC) in a nonlinear β-barium borate (BBO) crystal, pumped by a continuous-wave diode laser operating at a wavelength of 405 nm. Energy conservation in the down-conversion process results in pairs of photons at a wavelength of 810 nm. The two photons of each pair are spatially separated into two distinct optical paths: the verification channel and the measurement channel.

The verification channel serves as a temporal reference. It is directly coupled to a time-to-digital converter (TDC), which records the arrival times of detected photons with high temporal resolution.

The measurement channel contains both a state-preparation stage and a polarization-analysis stage. After state preparation, photons are analyzed using a polarizing beam splitter (PBS) aligned in the computational (Z) basis, corresponding to horizontal (H) and vertical (V) polarization states. An output port of the PBS is connected to a TDC, which records the arrival times of the filtered photons.

B. STATE PREPARATION

Photons propagating through the measurement channel first pass through a polarization state-preparation stage consisting of a half-wave plate (HWP) and a quarter-wave plate (QWP). By adjusting the orientation angles of these wave plates, allows measurements in different polarization bases to be performed using a fixed PBS, with the desired basis implemented through polarization rotations.

To generate specific input states, a polarizer and an additional QWP are placed at the beginning of the measurement channel. The polarizer selects a well-defined initial polarization state, while the QWP enables further transformations. Appropriate combinations of these optical elements allow controlled preparation of the required quantum states.

C. PREPARED STATES

Four different input states are considered in this experiment:

- Fully mixed state:
$$\rho = \frac{1}{2}\left(|H\rangle\langle H| + |V\rangle\langle V|\right)$$
- Diagonal linear polarization:
$$|D\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$$
- Right-circular polarization:
$$|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i\,|V\rangle)$$

- An unknown polarization state

The photon pairs produced by the BBO crystal are initially unpolarized, corresponding to a maximally mixed polarization state. By inserting a polarizer oriented along the diagonal basis, only photons in the $|D\rangle$ state are transmitted. Placing a QWP after the polarizer allows the diagonal state to be transformed into the right-circularly polarized state by setting the optical axis of the QWP to $\pm\pi/4$.

The unknown state is generated by randomly rotating the QWP, resulting in an arbitrary polarization state whose parameters are not known a priori.

D. WAVE-PLATE AND POLARIZER REFERENCE ANGLES

All rotation angles reported in this work are defined relative to experimentally determined reference positions (zeros) of the corresponding optical elements. These reference angles account for mechanical offsets.

For the polarization analysis stage in the measurement channel, the zero angles of the wave plates are defined as follows. The half-wave plate (HWP) used for implementing the measurement projections has a reference angle of

$$\theta^{(\mathrm{mes})}_{\mathrm{HWP},0} = 36^\circ,$$

while the corresponding quarter-wave plate (QWP) has a reference angle of

$$\theta^{(\mathrm{mes})}_{\mathrm{QWP},0} = 38^\circ.$$

All HWP and QWP angles listed in Table 1 are specified relative to these reference positions.

In the pump beam path, the polarization of the laser is controlled by a HWP whose reference angle is

$$\theta^{(\mathrm{pump})}_{\mathrm{HWP},0} = 105.5^\circ.$$

This wave plate is used to optimize the polarization of the pump laser incident on the BBO crystal, thereby maximizing the efficiency of the down-conversion process.

Finally, in the state-preparation stage of the measurement channel, the QWP placed after the polarizer has a reference angle of

$$\theta^{(\mathrm{prep})}_{\mathrm{QWP},0} = 105^\circ.$$

This reference defines the zero orientation of the QWP relative to the transmission axis of the polarizer and is used as the baseline for generating different input polarization states.

E. MEASUREMENT PROCEDURE

Since the PBS directly used for the measurement resolves only projections onto the $|H\rangle$ and $|V\rangle$ states, additional wave plates are placed before the PBS to enable measurements in other polarization bases. By applying suitable unitary rotations with a HWP and a QWP, arbitrary projective measurements can be mapped onto the fixed PBS basis.

The wave-plate relative angles used for the different projective measurements are summarized in Table 1.

TABLE 1. Wave-plate relative angles used for different projective measurements

Projection	HWP angle	QWP angle
$\|H\rangle\langle H\|$	0	0
$\|V\rangle\langle V\|$	$\pm\pi/4$	0
$\|D\rangle\langle D\|$	$+\pi/8$	0
$\|A\rangle\langle A\|$	$-\pi/8$	0
$\|R\rangle\langle R\|$	$-\pi/8$	$-\pi/4$
$\|L\rangle\langle L\|$	$+\pi/8$	$+\pi/4$

II. DATA ACQUISITION AND PREPROCESSING

A. DATA ACQUISITION

Single-photon detection events are recorded by the TDC and stored in plain-text (`.txt`) files. Each detection event is characterized by a time stamp and a channel identifier, indicating whether the photon was detected in the measurement channel (`tag: 3`) or in the verification channel (`tag: 2`). The temporal resolution of the TDC corresponds to a time-bin width of 80.955 ps per unit.

For each prepared input state and for each projective measurement setting, a separate data file is acquired. This ensures that data corresponding to different experimental configurations can be analyzed independently.

B. DATA PREPROCESSING AND COINCIDENCE ANALYSIS

The raw data consist of two independent sequences of time stamps, one from the verification channel and one from the measurement channel. In order to identify correlated photon pairs, a coincidence analysis is performed.

For each data file, the min time differences

$$\Delta t_i = \min_j |t^i_{\mathrm{meas}} - t^j_{\mathrm{ver}}| \quad \forall i \in \text{verification}$$

between each detection i in the measurement channel (t_{meas}) and the each detection j in the verification channel (t_{ver}) are calculated. These time differences are accumulated into a histogram, revealing a coincidence peak corresponding to true photon-pair detections, superimposed on a background originating from dark counts and accidental coincidences.

An appropriate coincidence time window is then selected around the peak, based on the variance of the gaussian constructed over the peak. This window is chosen to retain genuine coincidences while suppressing uncorrelated background events. Only events whose time differences fall within this window are counted as valid coincidences.

The final coincidence count obtained for each projection measurement serves as the experimental estimate of the corresponding projection probability. These counts are subsequently used for further data analysis, such as state reconstruction and comparison of theoretical QRNG protocol.

In this example (Figure 1), the coincidence peak is clearly visible near $\Delta t = 0$, and the selected time window captures the majority of genuine coincidences. Events falling outside this window are considered accidental and are excluded from the calculation of projection probabilities. This procedure is applied to all datasets corresponding to different input

states and measurement projections, ensuring a consistent and reliable determination of the coincidence counts used for subsequent analysis.

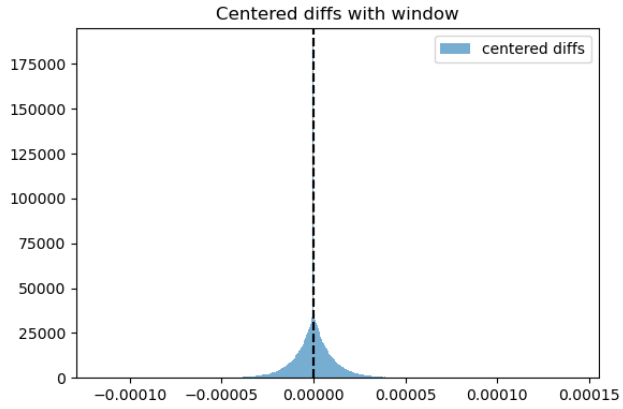


FIGURE 1. Example of a coincidence histogram obtained from a measurement run. (used time window: ± 1.42ns)

III. DATA ANALYSIS

The experimental data, analyzed by extracting coincidence counts from the measurement and verification channels can be used to evaluate the raw probability of each projection:

$$P^{\text{raw}}_{\text{projection}} = \frac{\text{\# measuring channel detection}}{\text{\# verification channel detection}}$$

filters through coincidence window, were used to calculate the probabilities of each projection:

$$P^{\text{filtered}}_{\text{projection}} = \frac{\text{\# filtered coincidence}}{\text{\# verification channel detection}}$$

1) Coincidence Probabilities

For each prepared state, raw and filtered probabilities counts were obtained for all six projection measurements: H, V, D, A, R, and L. The filtered probabilities were then normalized to compute the probability associated with each projection.

TABLE 2. Probability tables for the four input states.

State	Projection	Raw	Filtered	Normalized
Fully Mixed	H	0.389	0.039	0.468
	V	0.467	0.045	0.532
	D	0.436	0.045	0.508
	A	0.408	0.043	0.492
	R	0.408	0.041	0.491
	L	0.410	0.043	0.509
Diagonal	H	0.179	0.0171	0.541
	V	0.155	0.0145	0.459
	D	0.306	0.0315	0.997
	A	0.011	0.0001	0.003
	R	0.153	0.0152	0.473
	L	0.172	0.0169	0.527
Right-Circular	H	0.125	0.012	0.434
	V	0.158	0.015	0.566
	D	0.134	0.013	0.468
	A	0.151	0.015	0.531
	R	0.279	0.029	0.991
	L	0.013	0.000	0.009
Unknown	H	0.102	0.010	0.332
	V	0.197	0.020	0.668
	D	0.276	0.027	0.911
	A	0.035	0.003	0.089
	R	0.088	0.008	0.295
	L	0.193	0.019	0.705

2) Bloch Vector Reconstruction

The tomography of a projector can be used to represent a quantum state through measured expectation values. For a single qubit, any density matrix ρ can be expressed in the Bloch representation as

$$\rho = \frac{1}{2}\left(\mathbb{I} + \vec{r}\cdot\vec{\sigma}\right)$$

The norm $\|\vec{r}_{\text{proj}}\| \leq 1$ quantifies the purity retained after projection. Values close to unity indicate an effectively pure projected state, while smaller norms correspond to mixed projections.

The reconstructed block vector and it's norm for each considered state can be found in Table 3.

State	$r_x^{(\text{proj})}$	$r_y^{(\text{proj})}$	$r_z^{(\text{proj})}$	$\|\vec{r}_{\text{proj}}\|$
Mixed	0.0158	−0.0188	−0.0650	0.0695
Diagonal	0.9933	−0.0536	0.0821	0.9981
Right	−0.0633	0.9824	−0.1320	0.9933
Unknown	0.8214	−0.4098	−0.3365	0.9776

TABLE 3. Projected Bloch vectors and their norms for the four considered states

We can observe that:

- mixed state presents $\|\vec{r}_{\text{proj}}\| \approx 0$ coherently with a maximally mixed state.
- diagonal and right states have norms near the unit for σ_x and σ_y respectively.
- unknown state present a module near one coherently with it's preparation since it's a rotation of a pure state. Also the high component in all of the basis follows the idea of a random rotation.

3) Density Matrix Reconstruction

Given the boch vector, the corresponding density metrix can be calculated from: $\rho = \frac{1}{2}(\mathbb{I} + \vec{r} \cdot \vec{\sigma})$ The reconstructed density matrices are reported below:

TABLE 4. Reconstructed density matrices for the four input states.

State	Matrix	
Fully Mixed	0.467 0.008 - 0.009i	0.008 + 0.009i 0.532
Diagonal	0.541 0.497 - 0.027i	0.497 + 0.027i 0.459
Right-Circular	0.434 -0.032 + 0.491i	-0.032 - 0.492i 0.566
Unknown	0.332 0.4107 - 0.205i	0.411 + 0.205i 0.668

IV. QUANTUM RANDOMNESS EXTRACTION RESULTS

In this section we present the analysis of the different protocol based on the acquired data.

In particular each of them consider a different independence from the source.

A. PROTOCOL 1: TRUSTED QRNG

In the trusted-device scenario, both the source and the measurement apparatus are assumed to be fully characterized and reliable. Randomness is extracted from measurements performed in a single basis, and the min-entropy is directly estimated from the observed outcome probabilities.

$$H_{min} = -log_2(max\{p(+|i), p(-|i)\})$$

where i denotes the measurement base: $\{\sigma_x, \sigma_y, \sigma_z\}$. Since this protocol assume a trusted source the best min entropy is expected over state with maximum uncertain over the chosen base.

Here we report the result for all of the four state, considering all the possible base.

- **Mixed state**: $h_{\min}^{HV} = 0.909$ bits, $h_{\min}^{DA} = 0.977$ bits, $h_{\min}^{RL} = 0.973$ bits.
- **Diagonal state**: $h_{\min}^{HV} = 0.886$ bits, $h_{\min}^{DA} = 0.005$ bits, $h_{\min}^{RL} = 0.925$ bits.
- **Right state**: $h_{\min}^{HV} = 0.821$ bits, $h_{\min}^{DA} = 0.911$ bits, $h_{\min}^{RL} = 0.013$ bits.
- **Unknown state**: $h_{\min}^{HV} = 0.582$ bits, $h_{\min}^{DA} = 0.135$ bits, $h_{\min}^{RL} = 0.505$ bits.

As expected, the min-entropy favor the mixed state. We can also notice the entropy drop for the Diagonal and right in their respective polarized base while keeping a near maximal one in the other basis.

More interesting is the analysis over the unknown states that show the worst mean result

B. PROTOCOL 2: SDI-EUP

The Source-Device-Independent Entropic Uncertainty Principle (SDI-EUP) protocol relaxes the trust assumptions on the source while requiring measurements in two mutually unbiased bases. A Bound on the min-entropy is obtained from pairs of bases.

$$H_{min}(Z) \geq log_2 \frac{1}{c} - H_{max}(X)$$

Then since we are considering mutually unbiased bases $c = \frac{1}{d}$ where $d = 2$ for a qubit obtaining $H_{min}(Z) \geq 1 - H_{max}(X)$.

Here we report the result for all of the four state, always considering the measurement of HV for randomness while DA or RL for certification.

- **Mixed state**: $h_{\min}^{HV,DA} = 0.000$ bits, $h_{\min}^{HV,RL} = 0.000$ bits.
- **Diagonal state**: $h_{\min}^{HV,DA} = 0.842$ bits, $h_{\min}^{HV,RL} = 0.001$ bits.
- **Right state**: $h_{\min}^{HV,DA} = 0.001$ bits, $h_{\min}^{HV,RL} = 0.753$ bits.
- **Unknown state**: $h_{\min}^{HV,DA} = 0.349$ bits, $h_{\min}^{HV,RL} = 0.065$ bits.

In this case we can notice the $H_{min}(\text{mixed}) \approx 0$ proving the robustness of this bound against maximally mixed states.

The problems arise in case of a polarized state on a base that's not considered for verification. Infact we can notice from the knowledge of the whole state that it's safe to measure HV from Diagonal and Right state even if the protocol assume nearly zero randomness

C. PROTOCOL 3: SDI-TOMO

In the SDI-TOMO approach, measurements are performed in all three Pauli bases, allowing for a tomographic reconstruction of the effective Bloch vector $\mathbf{r}$. The min-entropy is then derived from:

$$\lambda_{max} = \frac{1 + |r|}{2}$$

. where $P_{guessing} \leq \lambda_m ax$, making it possible to calculate a bound of the min entropy: $H_{min} = -log_2(\lambda_{max})$.

Basically the protocol it's evaluating how pure the state is while making no assumption over the measuring basis. Infact highly pure states correspond to Bloch vectors with norm close to one, resulting in a strong reduction of certifiable randomness.

Here we report the result for all of the four state:

- **Mixed state**: $|\mathbf{r}| = 0.069$, $h_{\min} = 0.903$ bits.
- **Diagonal state**: $|\mathbf{r}| = 0.998$, $h_{\min} = 0.001$ bits.
- **Right-circular state**: $|\mathbf{r}| = 0.993$, $h_{\min} = 0.005$ bits.
- **Unknown state**: $|\mathbf{r}| = 0.978$, $h_{\min} = 0.016$ bits.

Even in this case we can notice how even if we now that the polarized state can be used for extracting randomness using the right base, the fact that the protocol doesn't consider the measuring basis cause them to be addressed as non secure.

D. PROTOCOL 4: SDI-POVM (OPTIONAL)

The SDI-POVM protocol exploits a four-outcome POVM to maximize randomness extraction in the SDI framework. This approach generally provides higher min-entropy compared to projective measurements. In particular we can define:

$F_k = \frac{1}{2}|k\rangle\langle k| \; \forall k \in \{H, V, D, A\}$ and compute $P_k^{raw} \propto Tr(|k\rangle\langle k|\,\rho)$ where rho is the mixed state density matrix. Finally:

$$H_{min} = -log_2(\max_k\{\frac{p_k}{p_H + p_V + p_D + p_A}\})$$

Here we report the result for all of the four state:

- **Mixed state**: $h_{\min} = 1.909$ bits.
- **Diagonal state**: $h_{\min} = 1.005$ bits.
- **Right state**: $h_{\min} = 1.821$ bits.
- **Unknown state**: $h_{\min} = 1.135$ bits.

V. SECURITY LEVEL ASSESSMENT

In this final assignment, the security of the considered system is evaluated for the **mixed state** and the **diagonal state** by analyzing the parameter ϵ using the **min-entropy bound from the SDI-TOMP protocol**. These two states were selected due to their contrasting security properties: the mixed state produces a high min-entropy, resulting in a higher level of security, while the diagonal state exhibits low min-entropy, corresponding to a lower security level.

The evaluation is performed in two ways:

1) Fixing the sequence length L and computing the resulting security parameter ε.
2) Fixing the desired security parameter ε and determining the required sequence length L.

The security of the extracted key is quantified using the **Leftover Hash Lemma (LHL)**, which provides a bound on ϵ in terms of the min-entropy of the state. For a system described by a state ρ_X, where X is the raw ke, the LHL reads:

$$\ell \leq H_{\min}^{block}(X) - 2\log\frac{1}{\varepsilon}, \tag{1}$$

where:

- ℓ is the length of the extracted secure key,
- $H_{\min}^{block}(X)$ is the min-entropy of X multiplied by the block size,
- ϵ_h is the failure probability of the randomness extraction,

Rewriting the LHL in terms of the security parameter for a fixed key length ℓ, one obtains:

$$\varepsilon = \frac{1}{2}2^{(\ell - H_{min}^{block}(X))} \tag{2}$$

which allows us to **compute ϵ for a fixed sequence length** L once the min-entropy is known. Conversely, solving for L in terms of a fixed ϵ yields the **required sequence length** to achieve a desired security level.

The results show a clear separation between the two states:

- For the **mixed state**, the high min-entropy ensures that ε rapidly decreases with increasing L, achieving strong security even for moderate sequence lengths.2
- For the **diagonal state**, the low min-entropy results and the fact that the length it's not fixed but a fraction of the block size makes the epsilon level to always increase. Also we can notice that for the same security parameter it's impossible to extract randomness since we should have a negative length.3

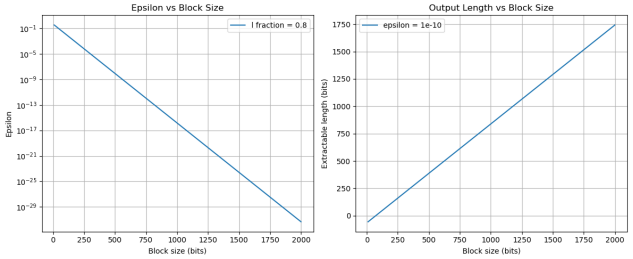


FIGURE 2. mixed state result, l fraction means that we use $\ell = 0.8 \cdot$ (block size)

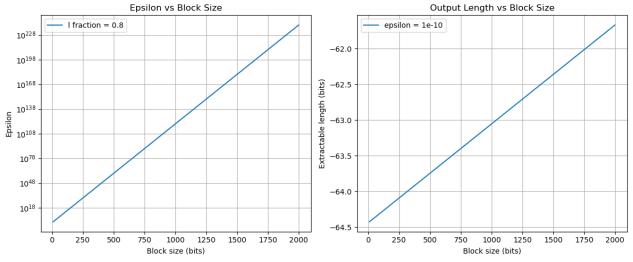


FIGURE 3. diagonal state result, l fraction means that we use $\ell = 0.8 \cdot$ (block size)

A. QRNG SIMULATION

As per final laboratory request a complete QRNG pipeline was implemented, using a Toeplitz matrix as universal2 seed extractor to follow the previous analysis assumption. In particular in the absence of simultaneous measurement over the complementary projection for a certain base, to keep the experiment as realistic as possible, we overlapped the two filtered measuring time frame to simulate a real word scenario. From that a raw string of bits was extracted and the randomness was extracted applying the Toeplitz matrix.